Four standard atoms in the current M_{11} subgroup table

L054 bounded recovery component

Source and acquired theorem. This component addresses only rows 1, 3, 4 and 6 of the retained current Connor–Leemans companion M_{11} table. Their native labels are respectively M_{11}, $L_2(11)$, A_6 and S_5. The numeric row bijection and actual saved subgroup generators are acquired from the accepted atlas computation. The printed structures are checked separately here; an order coincidence does not establish a group name.

We acquire Theorem MC.3 (Cayley transport) from the accepted project volume *Marked coordinates and exact orbit enumeration*, together with its distinct review and public admission. Its exact interface is this: for finite groups generated by ordered tuples (a_i) and (b_i), a bijection f with $f(1) = 1$ and

$$f(xa_i) = f(x)b_i \qquad \text{for every } x, i$$

is the unique isomorphism with those generator images. Every finite group is generated as a monoid by its group generators, so the positive-word induction in that acquired proof supplies full multiplicativity. We use this previously proved elementary result, with no novelty claim or new formalization. Checking all labelled edges costs $|G|$ times the number of generators, rather than checking all $|G|^2$ products.

Exact standard models. The A_6 model is the set of all even permutations of six points; its four listed 3-cycles generate exactly those 360 permutations by literal closure. The S_5 model consists of all 120 permutations on five points, with its listed 5-cycle and transposition. Thus these model identities use the definitions, not a small-group identifier or a structural-name oracle.

For $L_2(11) = \mathrm{PSL}_2(11)$, form all 2×2 matrices of determinant one over $\mathbb{F}_{11}$, and their fractional-linear actions on $\mathbb{F}_{11} \cup \{\infty\}$. A matrix fixing $0, 1, \infty$ is scalar; among determinant-one scalars it is $\pm I$. Consequently this literal image is the defining projective special linear group. The verifier forms all 1,320 matrices and all 660 distinct actions. It checks that the two specified elementary unipotent actions generate exactly that set. No classification of groups of order 660 is used.

The M_{11} model retains the already accepted named interface `MathieuGroup(11)` and its exact exported two permutations. The standard-name identification of this constructor is an explicit retained premise; it is not deduced merely from the computed order 7,920. The new verifier checks the literal constructor output against the acquired file, reconstructs its whole permutation closure and uses no GAP routine.

Theorem 1. *The actual retained subgroup representatives corresponding to the four source rows have the printed isomorphism types:*

source row	*actual record*	*order*	*generators*	*checked edges*
1 (M_{11})	39	7920	2	15840
3 ($L_2(11)$)	37	660	2	1320
4 (A_6)	36	360	4	1440
6 (S_5)	34	120	2	240

Proof. For each row the independent literal verifier parses the original saved generator array, checks its exact source-row join, and reconstructs the complete subgroup by finite positive-word closure. The shared registry contains the full literal standard model and its ordered generators. Its record is compared with the defining construction above. The supplied image vector is a bijection onto the whole actual subgroup, sends identity to identity, and satisfies every labelled right-generator edge. The total is 18,840 such checks. All hypotheses of acquired MC.3 hold, giving the four isomorphisms. The arguments apply to the actual representatives, not merely to groups of the same order. The four new registry models are indexed after the 29 acquired small models; the old registry is retained under its exact hash rather than altered. □

Evidence and limitations. Four isomorphism queries supplied the candidate maps in 2.38 seconds under a 120-second cap. GAP's `IsomorphismGroups` routine served only to find witnesses. The final checker uses neither GAP nor an isomorphism search, and independently checks every finite image. Ten damaged maps, generator arrays and source joins are rejected, also under Python optimization. The source native first page was directly inspected, including its subscripts and punctuation. Both the literal generators and their acquired numerical provenance are bound; the earlier local-class duplicate issue is handled by the separately accepted robust-weight interface and does not change these saved representatives.

Together with the accepted 24-label pilot and corrected 17-label extension, these four assertions would account for all nine A_5 labels and 36 of the 39 M_{11} labels after distinct acceptance. The remaining rows 2, 5 and 7 contain M_{10}, $M_9 : 2$ and M_9, whose precise naming conventions remain separate. No historical download identity, canonical name uniqueness, whole atlas, original-paper completion, conjecture, Lean or measurement credit is claimed. Root suggested registry reuse and the labelled-edge method; the current author found and acquires the existing complete MC.3 proof, and authored these literal models, joins and verifiers. Distinct review remains required.